EXPLAINABLE LIGHTWEIGHT AI ASSISTANT FOR DETECTING PRIVACY LEAKS IN SOCIAL MEDIA POSTS

Institute of Computer Science

UPLB

Natural Language Processing (NLP), Explainable AI (XAI), Data Privacy

Presented by: **Jesella Fermo**

Adviser: **Asst. Prof. Joseph Anthony C. Hermocilla**

Problem

The Privacy Gap

5.41B

Active Social Media users as of 2025

Privacy Paradox

Users care about privacy but rarely act on that concern

Cognitive Biases

Optimism bias & overconfidence impair risk perception

Tool Gap

Existing tools lack real-time, user-friendly explainability

What This Study Aims to Do

Objectives

1. Design a lightweight NLP model for privacy risk classification
2. Build an XAI component for non-technical, user-friendly explanations
3. Implement a browser extension that runs on resource-limited devices
4. Evaluate accuracy, latency, explainability & resource usage vs. baselines
5. Assess usability and trust via user studies with non-expert participants

Methodology

System Architecture

Privacy Detection Module

DistilBERT classifier fine-tuned for binary privacy risk classification

→

Explainability Module

SHAP computes token-level importance scores per prediction

→

Browser Extension UI

Chrome extension highlights risky tokens & shows risk level

Methodology

System Integration and Workflow

1 Input Detection
Content script finds textarea / contenteditable fields on the page

2 Text Capture
400 ms debounce → sends text only after typing stops

3 Backend Request
Service worker POSTs JSON payload to /predict endpoint

4 Tokenization and Inference
DistilBERT tokenizes → softmax → class probabilities

5 Risk Classification
0.5 threshold → Low / Medium / High Risk or Safe tiers

6 SHAP Explanation
Token importance scores ranked and paired with tokens

7 Response Transmitted
JSON returns: label, confidence, assessment level, tokens

8 Visual Feedback
Popup updates; highlighter overlays color-coded underlines

Methodology

Model & Dataset

Model

- Base: DistilBERT (Hugging Face)
- Task: Binary classification (risk / safe)
- Deployed via FastAPI (POST /predict)
- Confidence → human-readable risk level

Dataset

- Source: Mendeley – User Sensitive Data
- 78,916 labeled social media text samples
- Binary labels: sensitive / non-sensitive
- Split: 70% train · 15% val · 15% test

Results

Classification Performance

Model	Accuracy	Precision	Recall	F1-score
BERT (Baseline)	0.8966	0.90	0.90	0.90
DistilBERT	0.9067	0.91	0.91	0.91
DistilBERT+XAI	0.9067	0.91	0.91	0.91

DistilBERT > BERT

1% accuracy & F1 improvement ($p < 0.001$, McNemar's test)

XAI adds zero cost

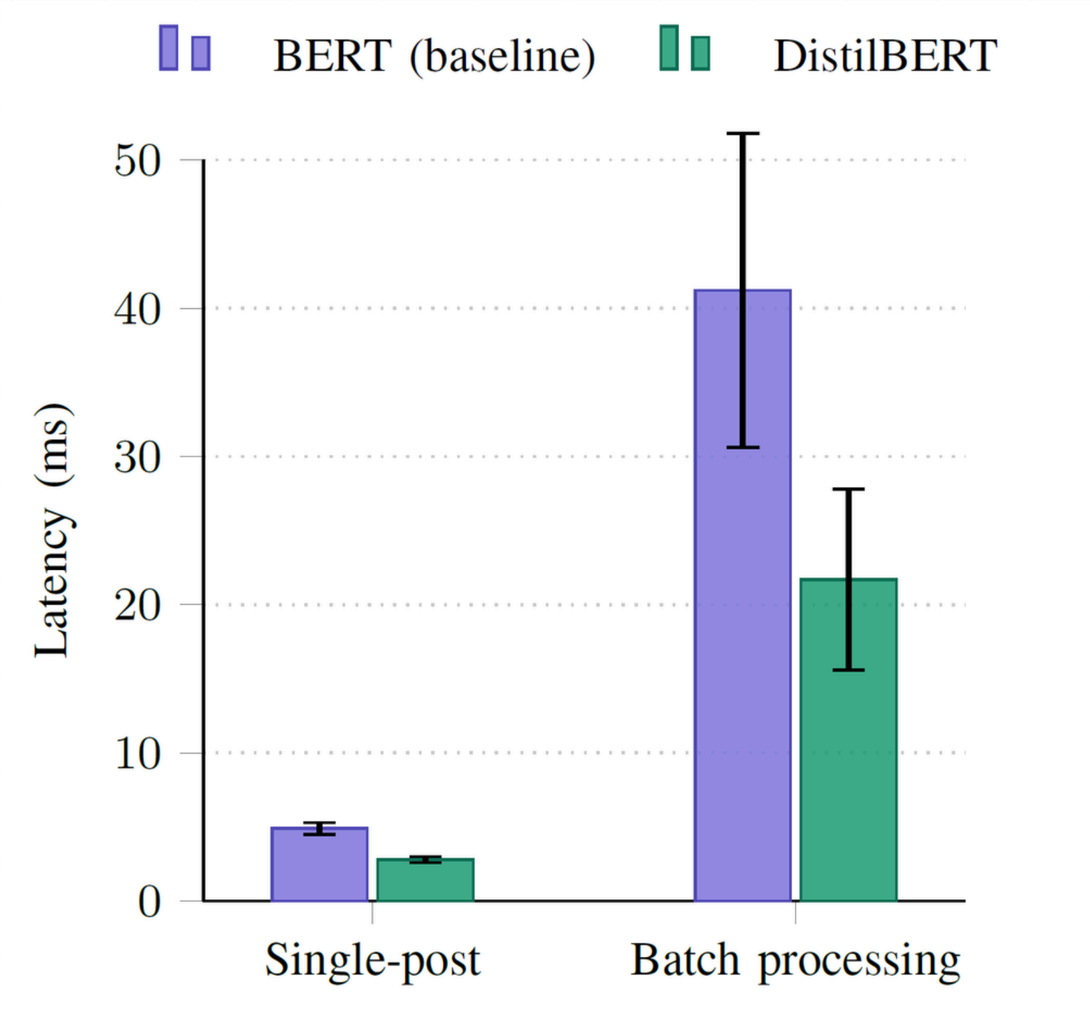
Explainability module does not affect classification scores

Contextual advantage

Transformer representations beat rule-based heuristics

Results

Efficiency & Scalability



High-End GPU

Latency (full test):	18.81s
RAM Usage:	621 MB
Accuracy:	90.67%

CPU-only

Latency (full test):	538.60s
RAM Usage:	303.46 MB
Accuracy:	90.67%

Results

Explainability and Robustness

XAI Latency Breakdown		
Operation	Mean	P95
Model Prediction	0.0242s	0.0262s
SHAP Explanation	0.5336s	1.0465s
Total Response	0.5616s	1.0703s
< 1s average total response time		

Robustness to Noisy Text			
Condition	Accurac	Precisio	Recall
Clean Text	90.67%	91%	91%
Noisy Text	88.17%	88%	88%
Only a 2.5% drop under noisy conditions, still reliable for social media text			

Results

User-Centered Evaluation

85.6

Mean SUS Score

Excellent Usability

9.16

Mean Recommendation Score (out of 10)

12

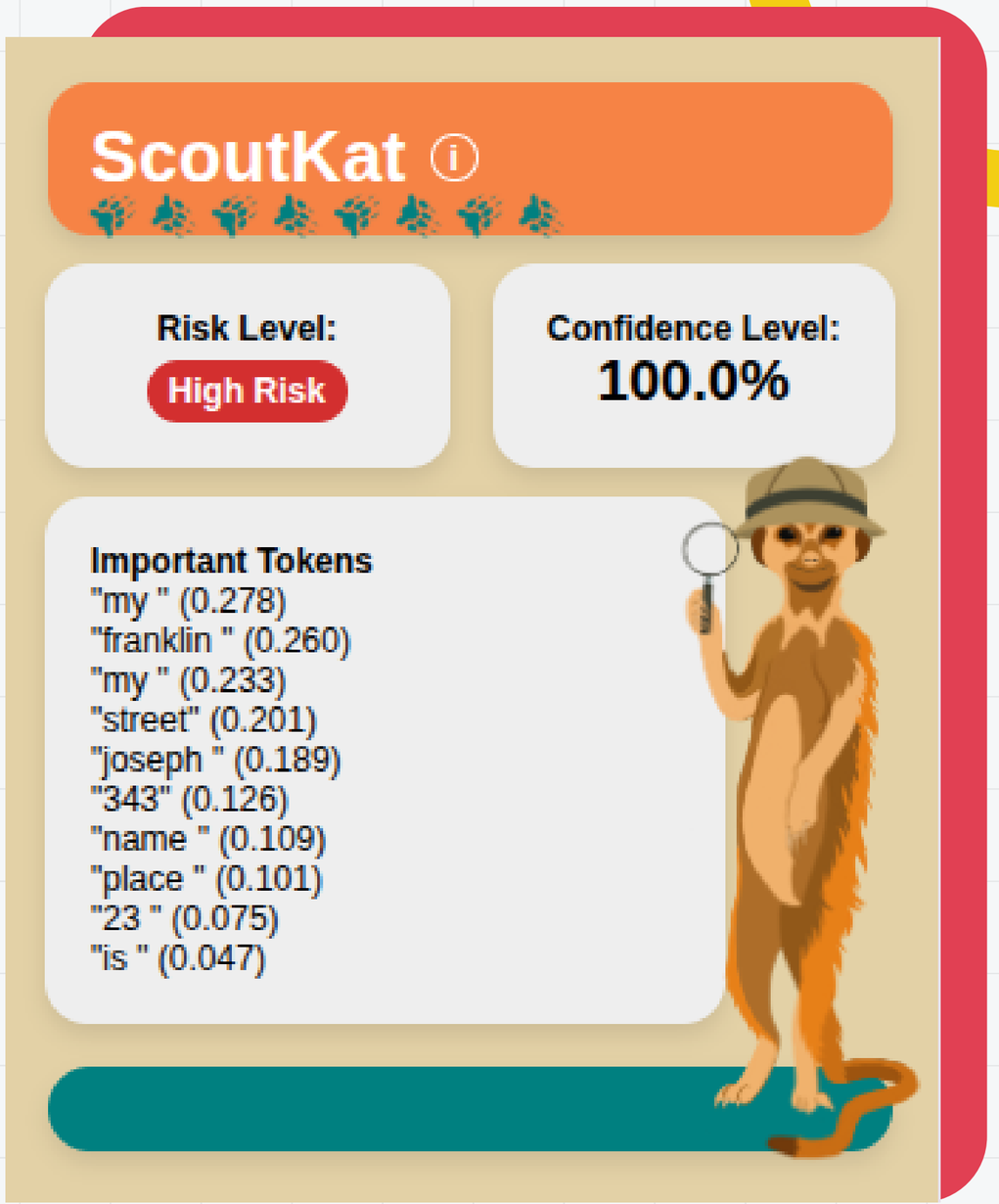
Non-expert Participants (screened, no privacy experts)

Prototype

Browser Extension Demo

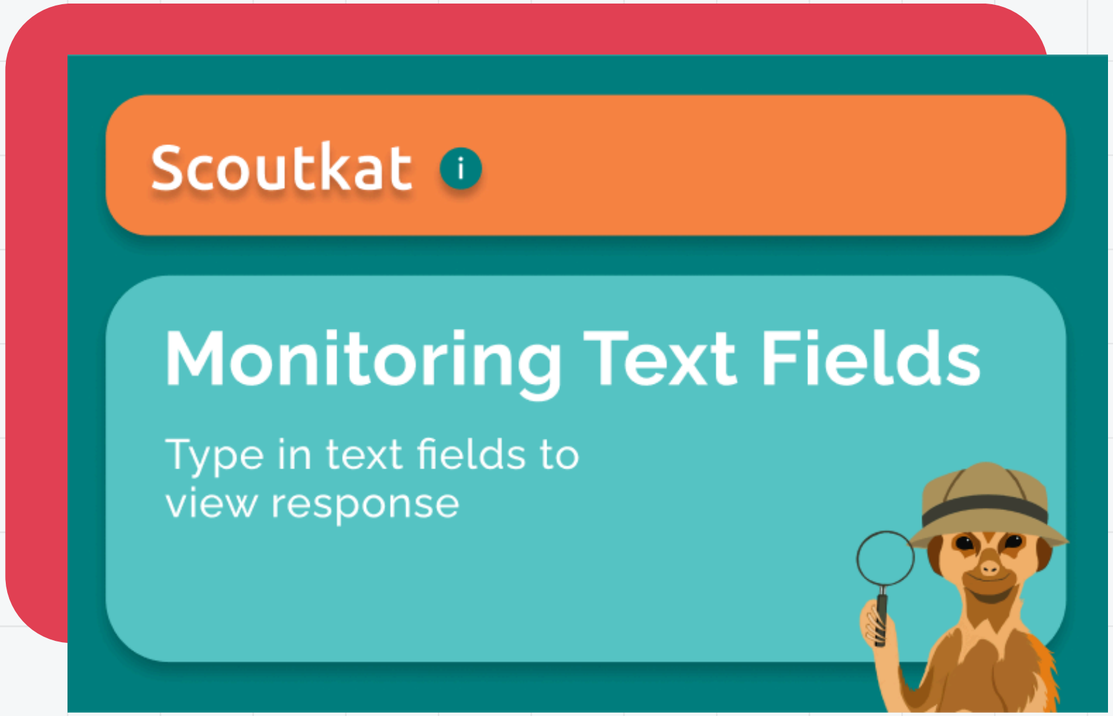
Main Interface

- Risk level (Low / Medium / High)
- Model confidence percentage
- Top contributing SHAP tokens
- Color-coded token highlights



Prototype

Browser Extension Demo



Loading Screen

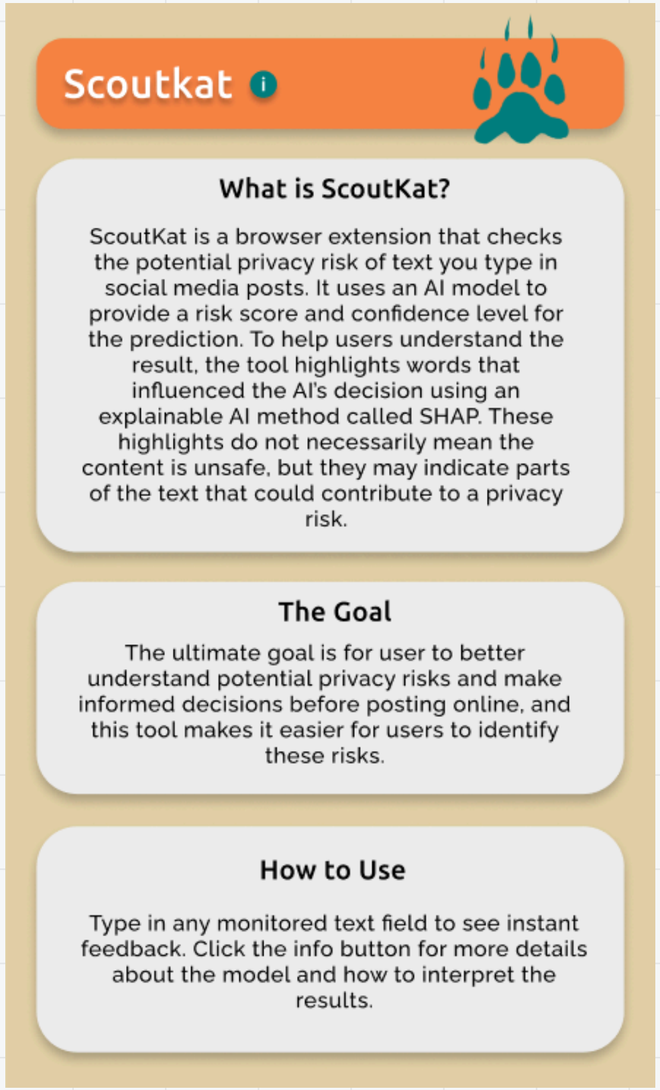
- Shown while processing request
- Shown when no text is detected
- Prevents premature user action

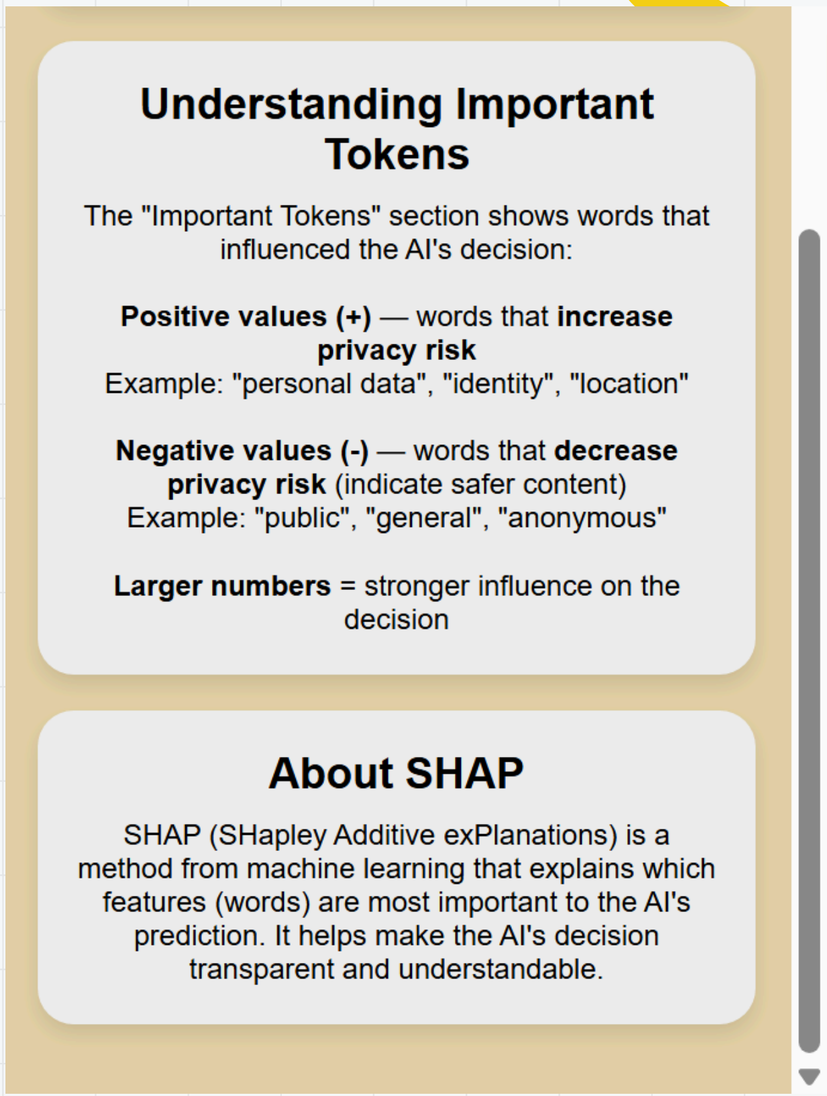
Prototype

Browser Extension Demo

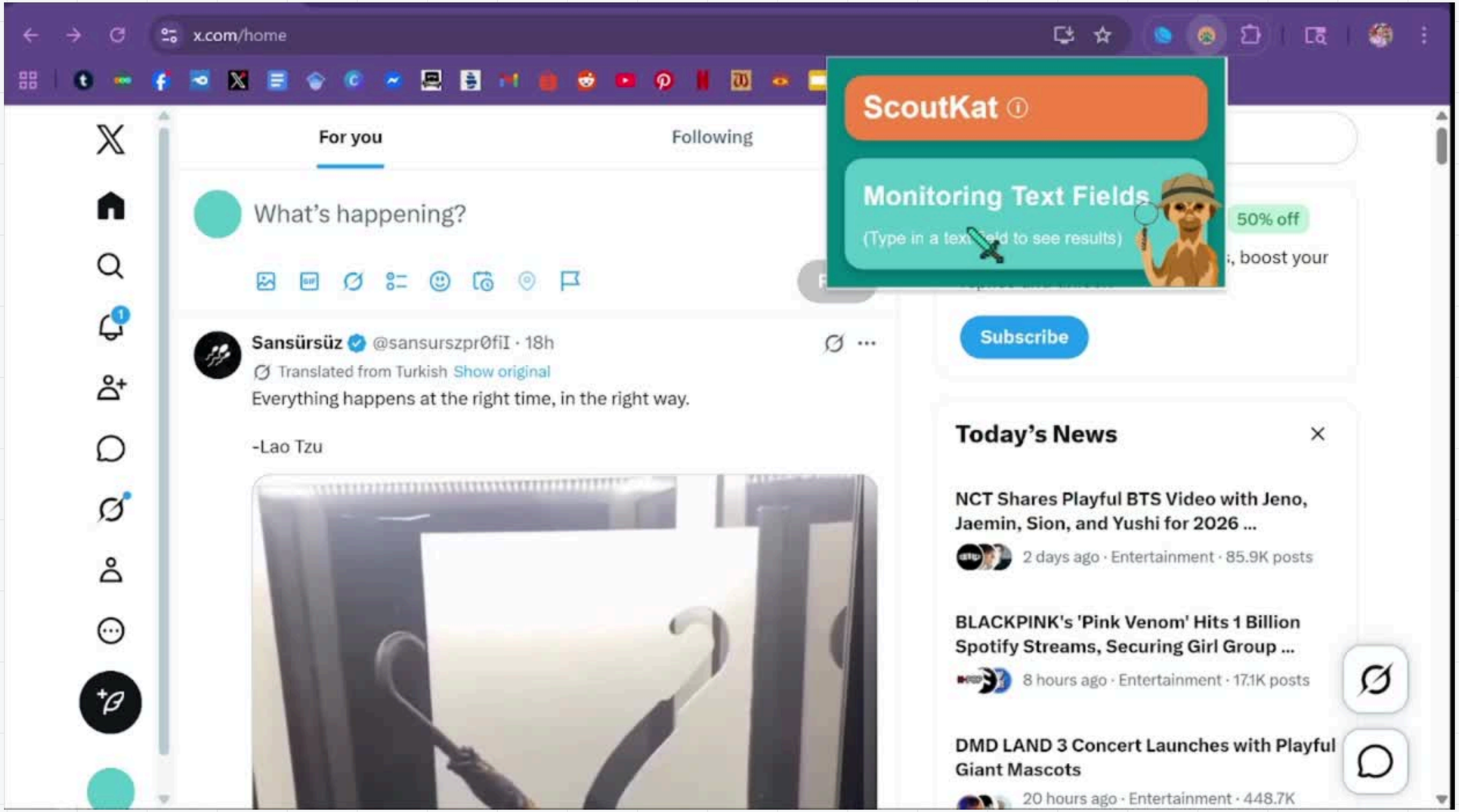
Info Page

- About / Goal / How-to-Use
- Visual guide to token colors
- SHAP explanation in plain language





Prototype Sample Run



Conclusion

- DistilBERT outperforms full BERT with 90.67% accuracy — statistically significant ($p < 0.001$)
- XAI (SHAP) adds explainability with < 1s average response — practical for real-time use
- System works across hardware environments — maintains accuracy regardless of GPU availability
- SUS score 85.6 and recommendation score 9.16 confirm non-expert accessibility and trust
- Addresses the 'privacy paradox' by turning passive concern into active, informed decision-making

Limitations and Future Work

- Reduce SHAP overhead
- Multilingual support
- Cloud deployment
- Multi-modal input

References

- Kepios, “Social media users,” https://datareportal.com/social-media-users, 2025, accessed: October 12, 2025.
- G. Canfora, A. Di Sorbo, E. Emanuele, S. Forootani, and C. A. Visaggio, “A nlp-based solution to prevent from privacy leaks in social network posts,” in Proceedings of the 13th International Conference on Availability, Reliability and Security, ser. ARES ’18. New York, NY, USA: Association for Computing Machinery, 2018. [Online]. Available: https://doi.org/10.1145/3230833.3230845
- V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter,” 2020. [Online]. Available: https://arxiv.org/abs/1910.01108
- S. Barth and M. D. de Jong, “The privacy paradox – investigating discrepancies between expressed privacy concerns and actual online behavior – a systematic literature review,” Telematics and Informatics, vol. 34, no. 7, pp. 1038–1058, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0736585317302022
- Kaddoura, “Dataset for user sensitive data on social media,” Mendeley Data, 2025.
- A. Barredo Arrieta, N. D´ıaz-Rodr´ıguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, “Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai,” Information Fusion, vol. 58, pp. 82–115, 2020. [Online].

ICS UPLB

Thank You For Listening

Any Questions?

Presented by: **Jesella Fermo**